

maguerero

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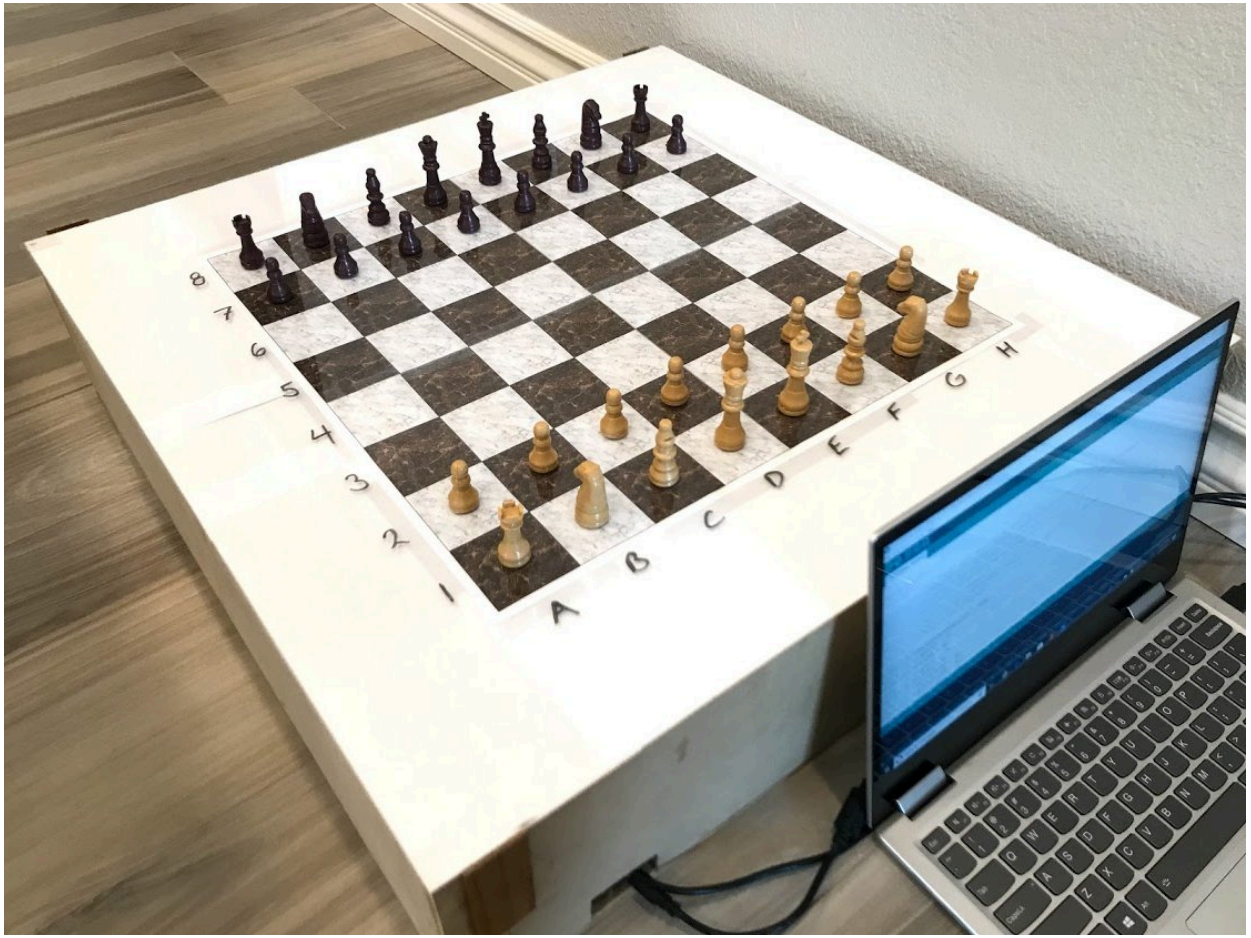
Automated Chess Board

Moves chess pieces around the board invisibly and seemingly magically.

Oct 24, 2018 • 87695 views • 91 respects

games robots

91 47



Devices & Components

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Arduino Uno Rev3

5	Linear Bearing Block

1	Medium Density Fiberboard	
1	Electromagnet	
1	Steel Rod Stock	
2	Capacitor 100 µF	
1	Acrylic Sheet	
1	Dual H-Bridge motor drivers L298	
2	NEMA 17 Stepper Motor	
2	Stepper motor driver board A4988	

91 47

2	Micro Limit Switch	

Hardware & Tools

1	Hacksaw
1	Power Drill

Project description

I wanted to make something cool while challenging my proficiency with Arduino and building electromechanical systems. I was interested in using magnets to do something to make it appear magical so I basically set out to make the ' *Wizard's Chess* ' game from the H between two people sharing the computer; the comp unfortunately.

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The board is hooked up to a computer and requires the player to input a square address of the piece to be moved and the new location for the piece. The electromagnet is moved by two stepper motors and timing belts. Each piece has a loose steel ball bearing inside to allow them to easily slide around.



Automated Chessboard Prototype Demo

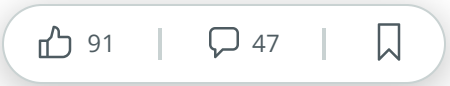
Mike's Makes

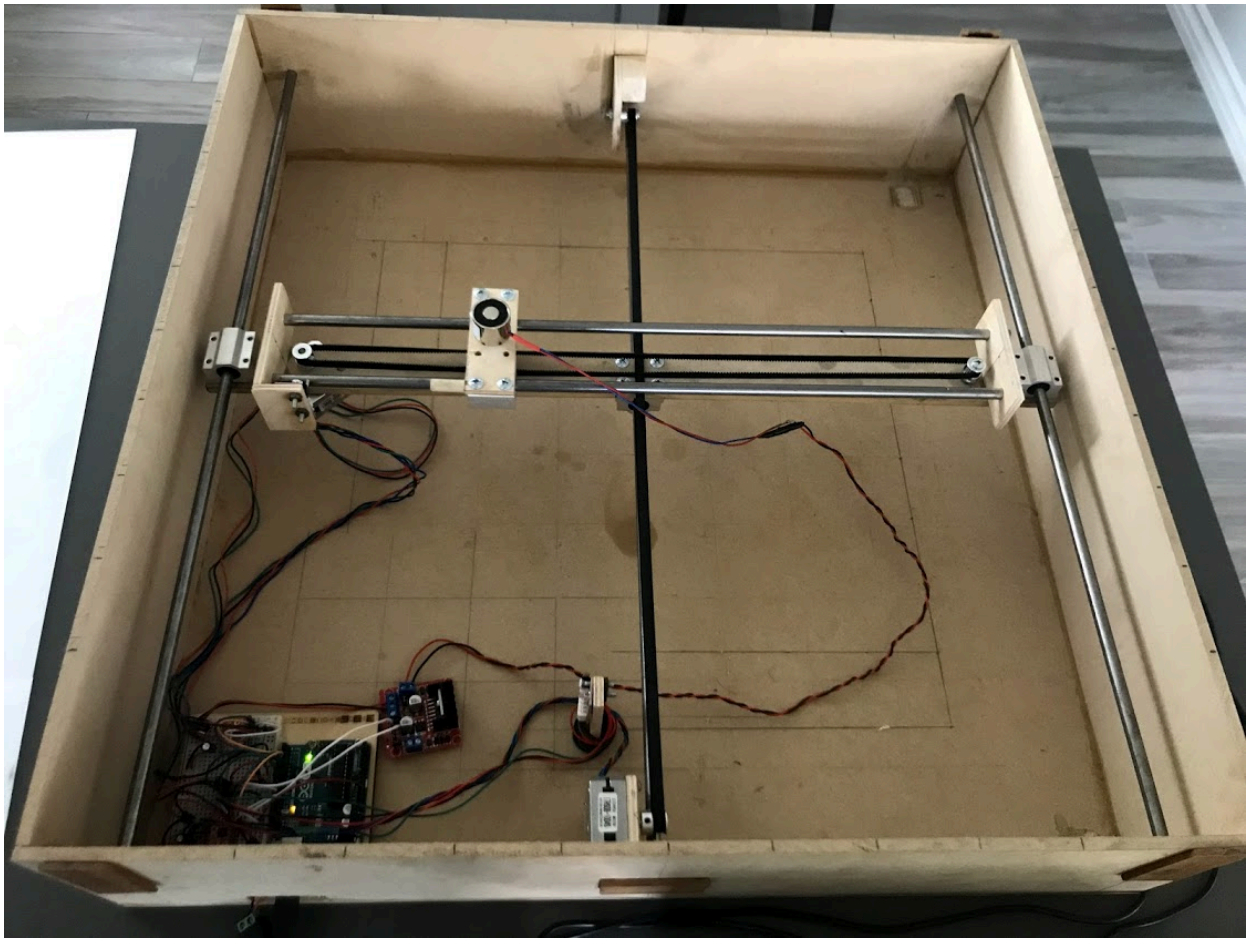






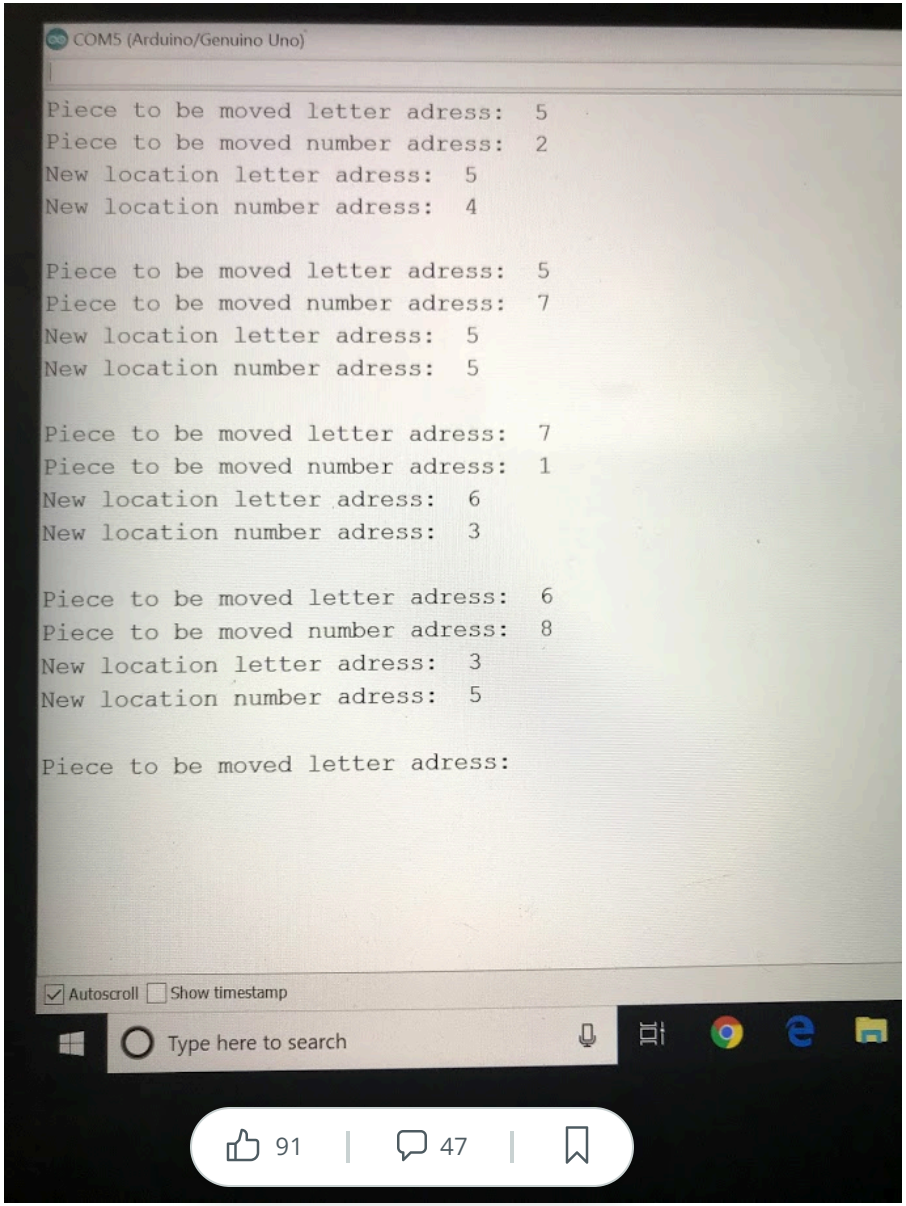
Watch on





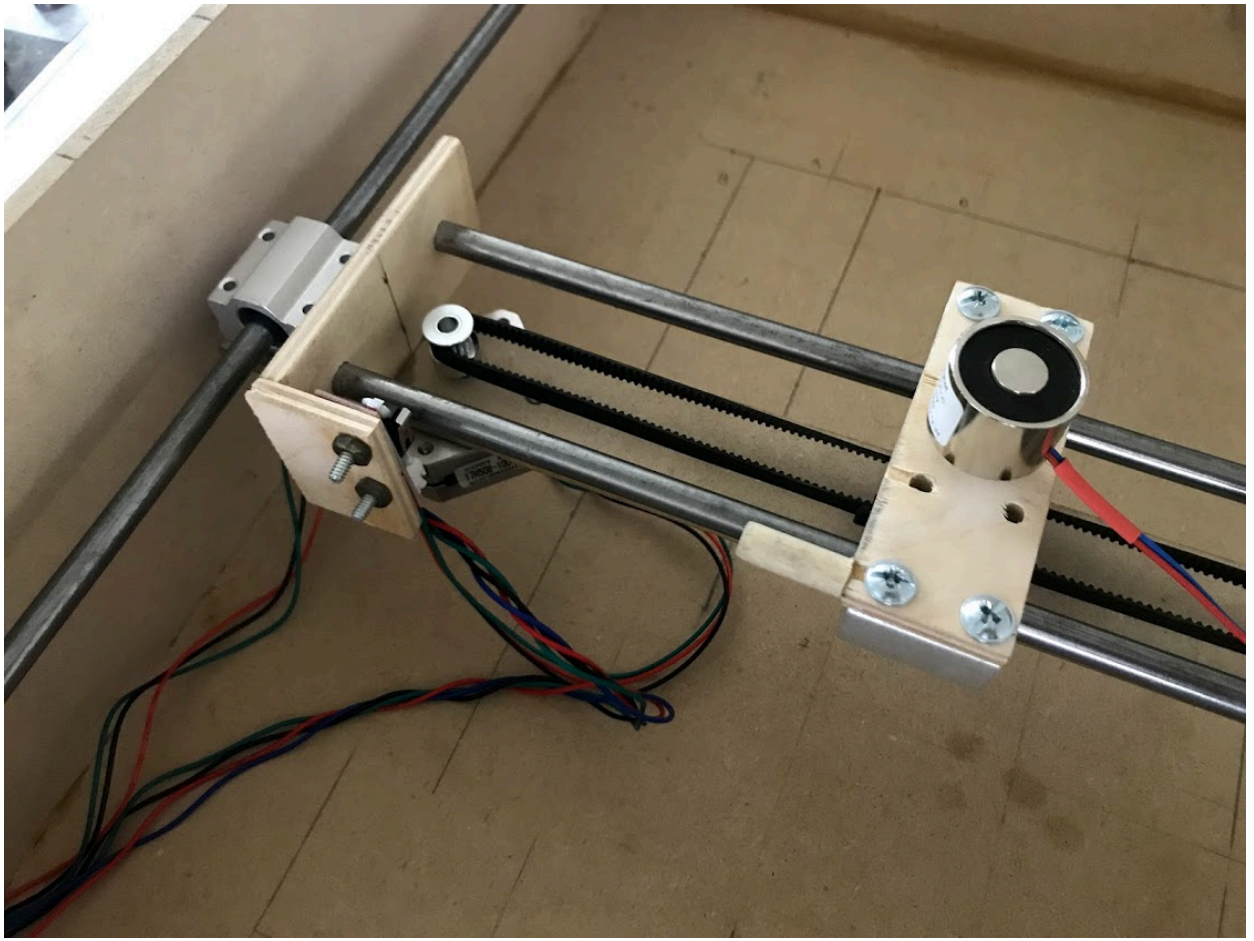
The biggest problem I had was getting the vertical axis to move smoothly without catching and missing steps. It doesn't happen often but when it does I just had to close the serial monitor and reopen it which reset the Arduino and zeroed the magnet pos.

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Serial monitor output for the demo video. Letter addresses are input as capital letters but reprinted as numbers.

Pieces are initially offset to the corner and moved along the lines between squares to avoid colliding with other pieces.



Featured as a Top Project in the December 2018 issue of HackSpace Magazine (https://hackspace.raspberrypi.org/)

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Code

Automated_chessboard arduino

Code will: *Home magnet with end stops at position A1. *Receive square address (Letter, Number) from Serial Monitor for piece to be moved. *Translate input into

number of steps to be completed by stepper motors. *Receive square address (Letter, Number) from Serial Monitor for new piece location. *Turn on electromagnet. *Translate input into number of steps to be completed by stepper motors. *Turn off electromagnet. Notes: *Stepper motors will run simultaneously (during zeroing and moving with the magnet off) until one reaches its max steps and then the other will continue until it reaches the specified square. *When the magnet is on and moving a piece the steppers initially offset to the corner and moves along the line between squares to prevent pieces interfering with each other. For example, this allows someone to move their knight before moving the pawns in front of it.

```
    delay(delayTime);
    digitalWrite(stepPinX, LOW);
    digitalWrite(stepPinY,  LOW);
    delay(delayTime);
}

////////////////////////////////////////////////////////////////
/////////////////////////////////////////

void  moveLinear(int stepPin, int delayTime){
  digitalWrite(stepPin, HIGH);
  delay(delayTime);
  digitalWrite(s
  delay(delayTime);
}
```

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Downloadable files

Automated_chessboard_Layout

Board powered from USB cable for all logic devices. Steppers and electromagnet (Solenoid) are powered from external power brick that is 9 volts and max 5 amps.

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Comments

Only logged in users can leave comments

LOGIN

M **masa22**

3 years ago

Hello, can you please tell me what is the borders of the squares?



zakariyea_

3 years ago

I love this project, it exactly what I was looking for. But I have a question: what happen exactly when a piece took (eat) another piece ?

W **waifupower**

4 years ago

Hello can i ask about the making and building of the project because I have a card time doing it. Can you show how u build the board?

91 | 47 |

A **Anonymous user**

4 years ago

Can you mention that which stepper motor is X and which is Y in the code by comparing the structure?

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Anonymous user

4 years ago

Hello!
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How can the string be modified?

```
Serial.print("Piece to be moved letter adress: ");
while (!Serial.available());
x1 = readSerialLetter();

Serial.print("Pi                    s: ");
while (!Serial.a
y1 = readSerialNumber();
```

91 | 47

Furthermore, if you are interested, I have solved the mechanical problems that you complain, and I also have the chess engine, and many photos.
Let me know
www.alvaropasseri.it



Mustafiz_mm

4 years ago

How to work on the automated car than is it similar like this ..I actually need to build an autonomous project. Anyone know where I can find such a project??

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wachtend op ee
dank bij voorb
m.v.g. remo

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4 years ago

Acrylic Sheet how much urgent...
?

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A **Anonymous user**
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Does this project use Big easy diver or easy diver to drive stepper motors? What is the use of Dual H-Bridge motor drivers L298? Please

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maguerero
4 years ago

Thanks! I'm not very familiar with implementing voice control but I did consider using it in a future version of this project. The best way to go about it would probably be using a WiFi or Bluetooth module and connecting to it with your phone or computer which already has voice recognition technology.

A **Anonymous user**

4 years ago

Love it. Thanks for sharing. I also love the games on this site http://bringoldsports.com/. Check it out.

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Thank you for your answer, I understand. But both your schematic and physical images use both A4988 and L298N. Please give me a detailed connection description, for example: STEP-Pin~5, DIR-Pin4. I am very interested in this project, but my personal level is limited, please help me, thank you very much!

91 | 47

Anonymous user

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In addition, the electromagnet does not have sufficient suction to attract debris because the article adsorbed by the electromagnet requires a thickness of 8 mm and the electromagnet does not have sufficient suction across the board. How did you solve this problem?



maguerero

4 years ago

I can't think of what the exact pin connections were for the stepper drivers and I disassembled it a long time ago. There are dozens of good tutorials out there for the A4988 drivers that I followed that explain everything way better than I ever could so go check some of those out.

I made this project with almost no background in electronics and using online forums and video tutorials for almost every compon[illegible] doing (and failing) is most of the fu[illegible])

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The electromagnet worked for me because I was powering it with 9 Volts instead of 5. Also, the most important thing is to make the distance between the magnet and metal ball as small as possible. I used an acrylic sheet that was less than 2 mm thick and had the magnet almost touching the bottom side of it.



maguerero

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I tried a bunch of different motor drivers but ended up using two A4988 drivers. I used cheap steppers that needed adjustable current limiting to work properly. I tried the L298N and L293D but they would get way too hot because there not rated for the Amp of current I needed for the steppers.

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It would have b 91 47 for the code used in this project

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Also i noticed that the position of the board seems incorrect where the white square should be in the bottom right according to a players position.

here is a reference:
https://www.chessvariants.com/fidelaws.html
1.2
The chessboard is placed between the players in such a way that the near corner to the right of each player is white.

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As for the Chess computer i think they used a Z80 CPU here is one that i found for inspiration :)
https://www.amazon.com/Sargon-Computer-Program-Dan-Spracklen/dp/0810451557



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Hi, what changes do you think would need to be made in order to make it connect to another board somewhere else so you could have a 1v1 match with someone somewhere else?

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atpd518

7 years ago

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